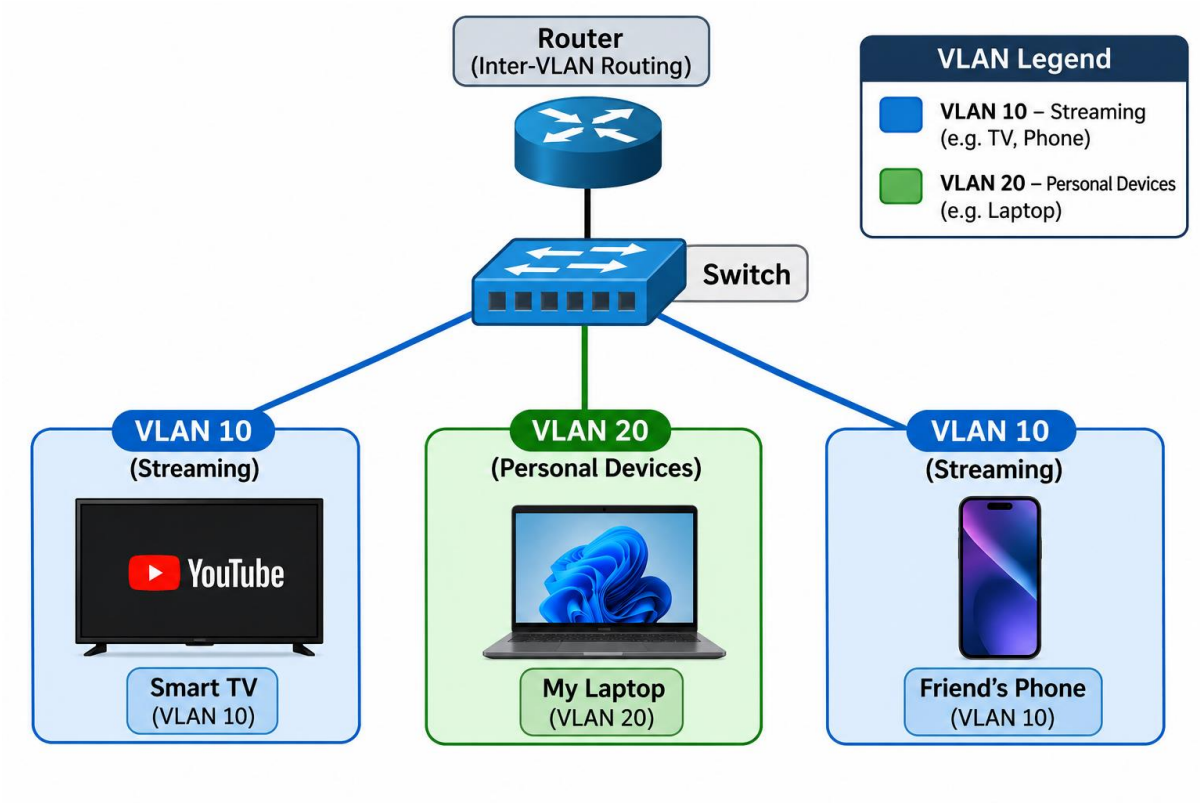
Switching and VLANS

Task 1:- Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices



Priyansh Bhaliya

Task 2:- Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other > Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI

Create Topology

- 1 Switch (2960)
- 5 PCs connected to switch

2. Assign IP Addresses

PC	IP Address	VLAN
PC0	192.168.10.1	10
PC1	192.168.10.2	10
PC2	192.168.10.3	10
PC3	192.168.20.1	20
PC4	192.168.20.2	20

Subnet Mask: `255.255.255.0`

3. Switch CLI Commands

```
enable
configure terminal
vlan 10
exit
vlan 20
exit
interface range fa0/1-3
switchport mode access
switchport access vlan 10
exit
interface range fa0/4-5
switchport mode access
switchport access vlan 20
exit
end
show vlan brief
```

4. Ping Test

- Same VLAN: PC0 → `ping 192.168.10.2` — Success
- Different VLAN: PC0 → `ping 192.168.20.1` — Fail

Task 3:- In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change

Before Configuration (PC0 in VLAN 10)

```
enable
configure terminal
interface fa0/1
switchport mode access
switchport access vlan 10
end
show vlan brief
```

2. Move PC0 to VLAN 20

```
enable
configure terminal
interface fa0/1
switchport access vlan 20
end
show vlan brief
```

3. Test Connectivity

- Set PC0 IP: `192.168.20.3`
- Set PC1 (VLAN 20) IP: `192.168.20.2`
- Subnet Mask: `255.255.255.0`

PC0 Command Prompt:

```
ping 192.168.20.2
```

Expected result: Ping succeeds if PC1 is in VLAN 20 and the switch ports are connected correctly.

Task 4:- Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3) Use a loop or range command if supported by the tool to avoid repeating the same command 10 times

Cisco Packet Tracer – Switch CLI Commands

```
enable
configure terminal
vlan 30
exit
interface range fa0/1-10
switchport mode access
switchport access vlan 30
end
show vlan brief
```

Result: Ports Fa0/1 to Fa0/10 are assigned to VLAN 30.